Lecture 6: Excessive Volatility in Prices and Returns

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Intro

Do Stock Prices Move Too Much to be Justified by Subsequent Changes in Dividends?

Motivation

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- r = constant discount rate
- With constant r, can this model generate the same volatility in prices as seen in the data?
- Price changes **must** be driven by news about future dividends!

Realized vs Forecasted Dividends

- The "efficient market model" implies:

$$P_t = \mathbb{E}_t\left[\sum_{k=1}^{\infty} \frac{D_{t+k}}{(1+r)^k}\right] = \sum_{k=1}^{\infty} \gamma^{k+1}\mathbb{E}_t[D_{t+k}], \qquad \gamma \equiv \frac{1}{1+r}$$

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- The realized series cannot be smoother than its projection:

$$\sigma^2(D_{t+k}) = \sigma^2(\mathbb{E}_t[D_{t+k}] + \varepsilon_{t+k}) = \sigma^2(\mathbb{E}_t[D_{t+k}]) + \sigma^2(\varepsilon_{t+k}) \geq \sigma^2(\mathbb{E}_t[D_{t+k}])$$

Shiller's Key Insight

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- Let Π_t be the GDP deflator and $A \equiv 1 + g$, where g a growth rate for real dividends
- Shiller deflated the index prices and dividends by inflation and the growth rate:

$$p_t \equiv \frac{P_t}{\Pi_t A^{t-T}}, \qquad d_t \equiv \frac{D_t}{\Pi_t A^{t+1-T}}$$

where $T = 1978$ is the last year of the sample and g is estimated as the constant log-growth rate of real dividends.

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- Similarly: $p_t^* = \sum_{k=1}^{\infty} \tilde{\gamma}^{k+1} d_{t+k}$

Shiller's Key Insight

- Crucially, we have the relationship:

$$p_t = \mathbb{E}_t[p_t^*] \implies \sigma^2(p_t) = \sigma^2(\mathbb{E}_t[p_t^*]) \leq \sigma^2(p_t^*)$$

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- Shiller assumed that they would grow at rate g forever
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- This assumption likely understates $\sigma^2(p_t^*)$
- p_{1978}^* is computed with "fake dividends" and then you use the recursion backward:

$$p_t^* = \tilde{\gamma}(d_{t+1} + p_{t+1}^*)$$

- The discount rate r is such that $r = \frac{\mathbb{E}[d]}{\mathbb{E}[p]}$ (why was deflating necessary for this step?)

The Main Plot

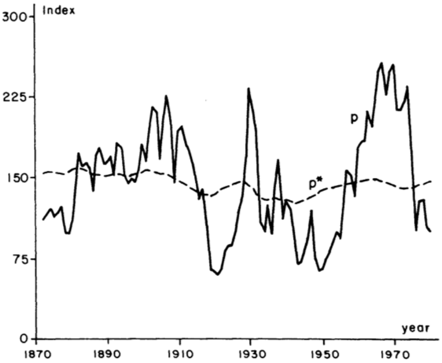


FIGURE 1

Note: Real Standard and Poor's Composite Stock Price Index (solid line p) and *ex post* rational price (dotted line p^*), 1871–1979, both detrended by dividing a long-run exponential growth factor. The variable p^* is the present value of actual subsequent real detrended dividends, subject to an assumption about the present value in 1979 of dividends thereafter. Data are from Data Set 1, Appendix.

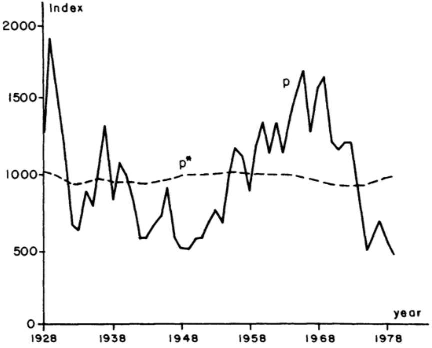


FIGURE 2

Note: Real modified Dow Jones Industrial Average (solid line p) and *ex post* rational price (dotted line p^*), 1928-1979, both detrended by dividing by a long-run exponential growth factor. The variable p^* is the present value of actual subsequent real detrended dividends, subject to an assumption about the present value in 1979 of dividends thereafter. Data are from Data Set 2, Appendix.

Main Numerical Result

- Strange things: inequalities upside down!
- You see many rows because he derived many bounds with different assumptions
- The data is rejecting theory by an order of magnitude
- $50 > 8$ and $355.9 > 26.8$

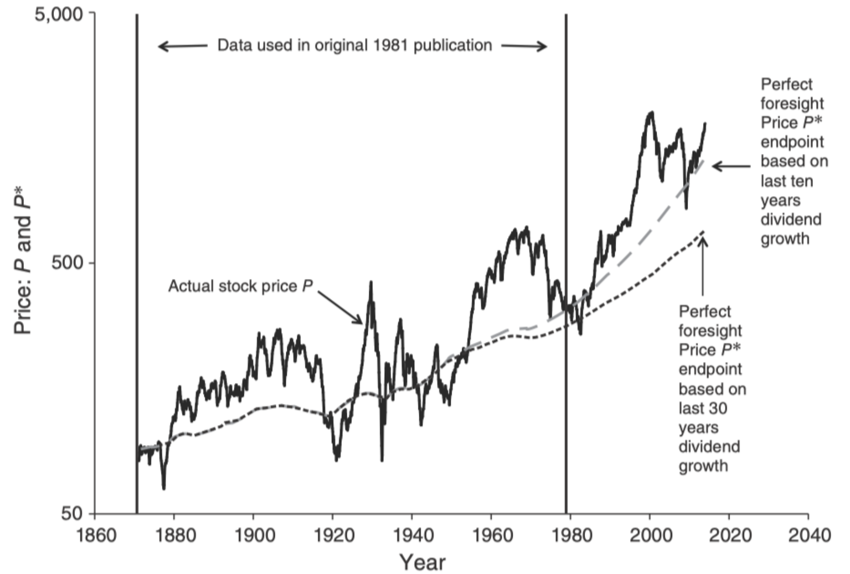
Historical evidence:

- During the Great Depression, (real) dividends were below average only on 1933-1935, and 1939
- But returns were pretty bad!

TABLE 2—SAMPLE STATISTICS FOR PRICE AND DIVIDEND SERIES

		Data Set 1: Standard and Poor's	Data Set 2: Modified Dow Industrial
	Sample Period:	1871–1979	1928–1979
1)	$E(p)$	145.5	982.6
	$E(d)$	6.989	44.76
2)	$\bar{r}$	.0480	0.456
	$\bar{r}_2$	.0984	.0932
3)	$b = ln\lambda$	.0148	.0188
	$\hat{\sigma}(b)$	(.0011)	(1.0035)
4)	$cor(p, p^*)$	.3918	.1626
	$\sigma(d)$	1.481	9.828
Elements of Inequalities:			
Inequality (1)			
5)	$\sigma(p)$	50.12	355.9
6)	$\sigma(p^*)$	8.968	26.80
Inequality (11)			
7)	$\sigma(\Delta p + d_{-1} - \bar{r}p_{-1})$	25.57	242.1
	$min(\sigma)$	23.01	209.0
8)	$\sigma(d)/\sqrt{\bar{r}_2}$	4.721	32.20
Inequality (13)			
9)	$\sigma(\Delta p)$	25.24	239.5
	$min(\sigma)$	22.71	206.4
10)	$\sigma(d)/\sqrt{2\bar{r}}$	4.777	32.56

Newer Data


5,000
500
50
Price: P and P*
Data used in original 1981 publication
Actual stock price P
Perfect foresight Price P* endpoint based on last ten years dividend growth
Perfect foresight Price P* endpoint based on last 30 years dividend growth
1860
1880
1900
1920
1940
1960
1980
2000
2020
2040
Year

What if Discount Rates Vary?

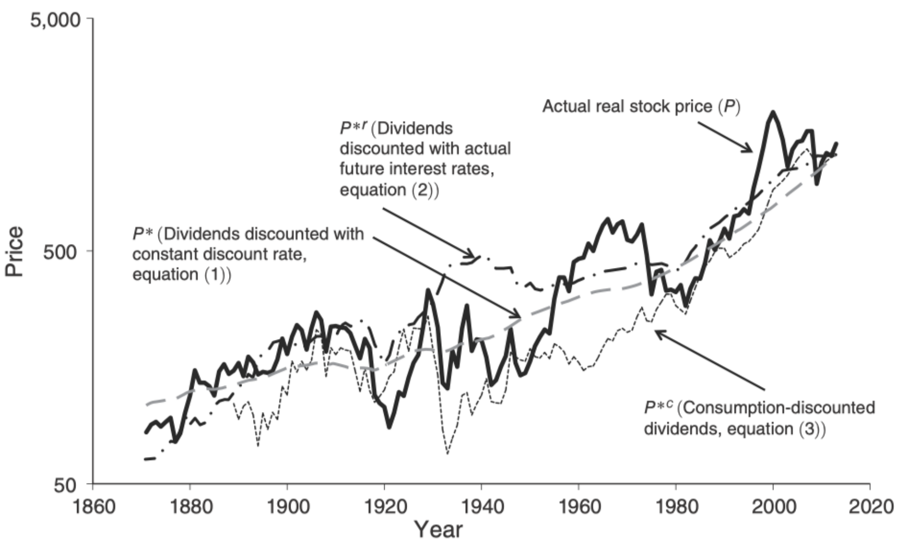
- We can always write:

$$P_t = \mathbb{E}_t \left[\sum_{k=1}^{\infty} D_{t+k} \prod_{j=1}^{k} \frac{1}{1 + r_{t+j-1}} \right]$$

- If we impose no discipline, this will always be true for some series of returns
- Shiller (1981) used 6-month prime commercial paper + a constant
- Shiller (2014) used our good and old friend:

$$r_{t+1} = -\log(\beta) + \delta \cdot \Delta c_{t+1}$$

Newer Data and Time-Varying Interest Rates



Where The Literature Took This

Takeaway message:

- Stock prices move too much to be justified by subsequent changes in dividends
- The natural way to reconcile this is allowing for rich time-variation in discount rates

Two Paths Forward:

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1. **Return decompositions**: how much of price movements is due to changes in dividend expectations vs expected returns? (Campbell and Shiller 1988; Campbell 1991; Cochrane 2011)

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Two Paths Forward:

1. **Return decompositions**: how much of price movements is due to changes in dividend expectations vs expected returns? (Campbell and Shiller 1988; Campbell 1991; Cochrane 2011)
2. **Behavioral finance and limits to arbitrage**: prices deviate from fundamentals due to investor psychology and limits to arbitrage (De Long et al. 1990; Shleifer and Vishny 1997; Shiller 2014; Barberis and Thaler 2003; Barberis 2018)

Behavioral Finance

- Excess volatility could reflect investor psychology, not rational risk premia
- Noise traders and limits to arbitrage (De Long et al. 1990; Shleifer and Vishny 1997)
- Shiller (2014): irrational exuberance and social dynamics drive speculative prices

Behavioral Finance

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- Noise traders and limits to arbitrage (De Long et al. 1990; Shleifer and Vishny 1997)
- Shiller (2014): irrational exuberance and social dynamics drive speculative prices
- Surveys: Barberis and Thaler (2003) and Barberis (2018) (extrapolation, overconfidence, prospect theory)
- We will not pursue this path, but it is an important and active research program

Questions?

Campbell-Shiller Decomposition

- Recall the log-linearization of prices and returns from Lecture 2:

$$p_t - d_t \approx k + \rho(p_{t+1} - d_{t+1}) + (1-\rho)r_{t+1} - d_{t+1}$$

- Iterating forward:

$$p_t - d_t \approx k + \sum_{j=1}^{\infty} \rho^{j-1} \left((1-\rho)r_{t+j} - \Delta d_{t+j} \right)$$

- This expression decomposes the log price-dividend ratio into expected future returns and expected future dividend growth
- We can use this to quantify how much of price movements are due to changes in expected returns vs expected dividend growth

AI-Generated Summary

Readings and References i

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